BIG DATA ENGINEERING MASTER INTERVIEW QUESTION BANK (100+ QUESTIONS)

1. Define Big Data.
2. Explain 5 V's of Big Data.
3. OLTP vs OLAP.
4. What is CAP theorem?
5. What is horizontal scaling?
6. What is vertical scaling?
7. Batch vs stream processing.
8. Structured vs unstructured data.
9. What is distributed computing?
10. What is cluster computing?
11. HDFS architecture explanation.
12. What is NameNode?
13. What is DataNode?
14. What is Secondary NameNode?
15. What is block size in HDFS?
16. What is replication factor?
17. What is rack awareness?
18. What is heartbeat in HDFS?
19. What is block report?
20. What is Safe Mode?
21. Explain HDFS write process.
22. Explain HDFS read process.
23. What is small file problem?
24. What happens if DataNode fails?
25. What happens if NameNode fails?
26. MapReduce architecture.
27. What is mapper?
28. What is reducer?
29. What is shuffle and sort?
30. What is combiner?
31. What is partitioner?
32. How many reducers to choose?
33. What is data skew?
34. Why Spark faster than MapReduce?
35. What is RDD?
36. What is DataFrame?
37. What is Dataset?
38. What is lazy evaluation?
39. What is DAG?
40. What is lineage?
41. Transformations vs actions.
42. Narrow vs wide transformations.
43. What is partitioning in Spark?
44. Repartition vs coalesce.
45. What is broadcast join?
46. What is shuffle join?
47. What is Spark driver?
48. What is executor?

49. What is SparkSession?
50. What is Catalyst optimizer?
51. What is Tungsten engine?
52. What is caching in Spark?
53. Cache vs persist.
54. What is checkpointing?
55. What is speculative execution?
56. Spark job optimization techniques.
57. How to handle data skew in Spark?
58. What causes shuffle?
59. What is window function in Spark?
60. What is UDF?
61. What is Kafka?
62. Kafka architecture.
63. What is topic?
64. What is partition?
65. What is offset?
66. What is consumer group?
67. What is replication in Kafka?
68. What is ISR?
69. At-least-once vs exactly-once.
70. What happens if broker fails?
71. How to increase Kafka throughput?
72. What is Hive?
73. Hive architecture.
74. Managed vs external tables.
75. What is partitioning in Hive?
76. What is bucketing?
77. ORC vs Parquet.
78. What is metastore?
79. How Hive query executed?
80. What is Data Lake?
81. What is Data Warehouse?
82. Schema-on-read vs schema-on-write.
83. What is ETL?
84. What is ELT?
85. What is Delta Lake?
86. What is Lakehouse architecture?
87. Design log analytics system.
88. Design fraud detection pipeline.
89. Design recommendation system.
90. Batch and streaming hybrid architecture.
91. How to troubleshoot slow Spark job?
92. Executor out-of-memory issue handling.
93. Kafka lag troubleshooting.
94. HDFS disk full scenario handling.
95. Handling small files in HDFS.
96. Why repartition(1) dangerous?
97. Why collect() dangerous?
98. Why too many partitions bad?
99. Why Spark job fails in cluster but works locally?

100. How to tune Spark configuration?
101. What is YARN?
102. What is resource manager?
103. What is driver memory vs executor memory?
104. What is dynamic allocation?
105. How to monitor Big Data pipeline?
106. What is Airflow?
107. What is data lineage?
108. What is CDC?
109. What is data governance?
110. What is ACID in Delta Lake?
111. What is watermarking in streaming?
112. What is structured streaming?
113. What is schema evolution?
114. What is data compaction?
115. What is indexing in Hive?
116. What is cost-based optimizer?
117. What is vectorized execution?
118. What is Bloom filter?
119. What is distributed join strategy?
120. End-to-end Big Data pipeline design.